

YOUR OFE 2025 REPORT

AUTHOR

Farmers DataLab

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2025 Cropping Season

Field 302-3 | Fixed rates: 25–62 lb N/ac | Variable-rate: R0, R1, R6, R9 | **Research question:** Do cover crops and improved soil health compensate for reductions in applied nitrogen when it comes to grain yield?

Welcome to your 2025 farm report. This report summarizes the on-farm nitrogen rate experiment conducted in field 302-3. This season, your entire field had cover crop, so the question we're asking is a step further than most: given that the cover crop is already in place, how much additional nitrogen fertilizer does the corn actually need? We tested five N rate scenarios across 10 strips (R0–R9): fixed rates of 25, 37, 50, and 62 lb N/ac, plus variable-rate strips (R0, R1, R6, R9) that averaged around 30 lb N/ac. Variable-rate strips are presented as a single "Variable" group, allowing a direct comparison of fixed vs. variable-rate N management. The goal is to find the rate where you get the most grain for your dollar, and see how far the cover crop can take you before yield starts to slip.

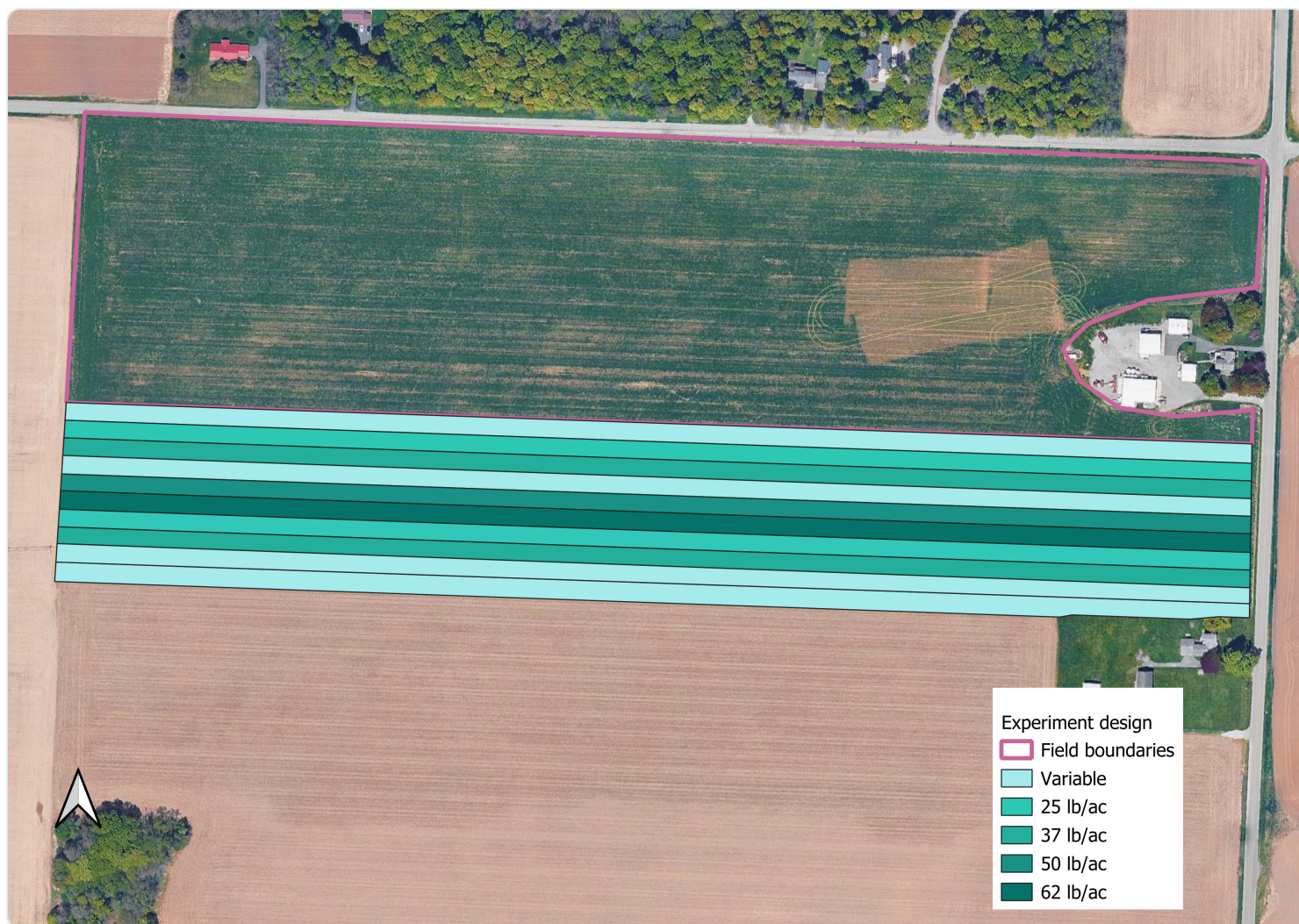


Figure 1. Field 302-3 experiment layout.

Take-home message: More nitrogen generally meant more nitrogen in the plant and higher stalk nitrate at season end, but it didn't always mean more yield. The 50 lb N/ac strip actually outperformed the 62 lb N/ac strip, suggesting there's a point where adding more fertilizer stops paying off. The variable-rate strips had the lowest average yield and stalk nitrate values just barely above the sufficiency threshold, although the low yield comes mostly from one strip on the southern edge of the field. Your soil chemistry is excellent, but biological activity is still developing. As the cover crop system matures, the soil should be able to supply more nitrogen internally, and the optimal fertilizer rate may shift lower over time.

Results Summary

N Rate Group	Plots	Mean %N (Biomass)	Mean CSNT (ppm)	Mean Yield (bu/ac)
25 lb N/ac	2	3.568	336	160.0
37 lb N/ac	2	3.868	419	165.5
50 lb N/ac	1	3.630	832	178.1
62 lb N/ac	1	4.167	943	170.8
Variable	4	3.628	257	155.7

Fixed N rate strips: R2 & R7 = 37 lb N/ac, R3 & R8 = 25 lb N/ac, R4 = 62 lb N/ac, R5 = 50 lb N/ac. Variable-rate strips: R0, R1, R6, R9. Small sample sizes (n = 1–4 plots per group) preclude formal statistical testing.

Corn Yield

The map below shows the **corn yield spatial distribution** across field 302-3, with the experimental strip design overlaid.

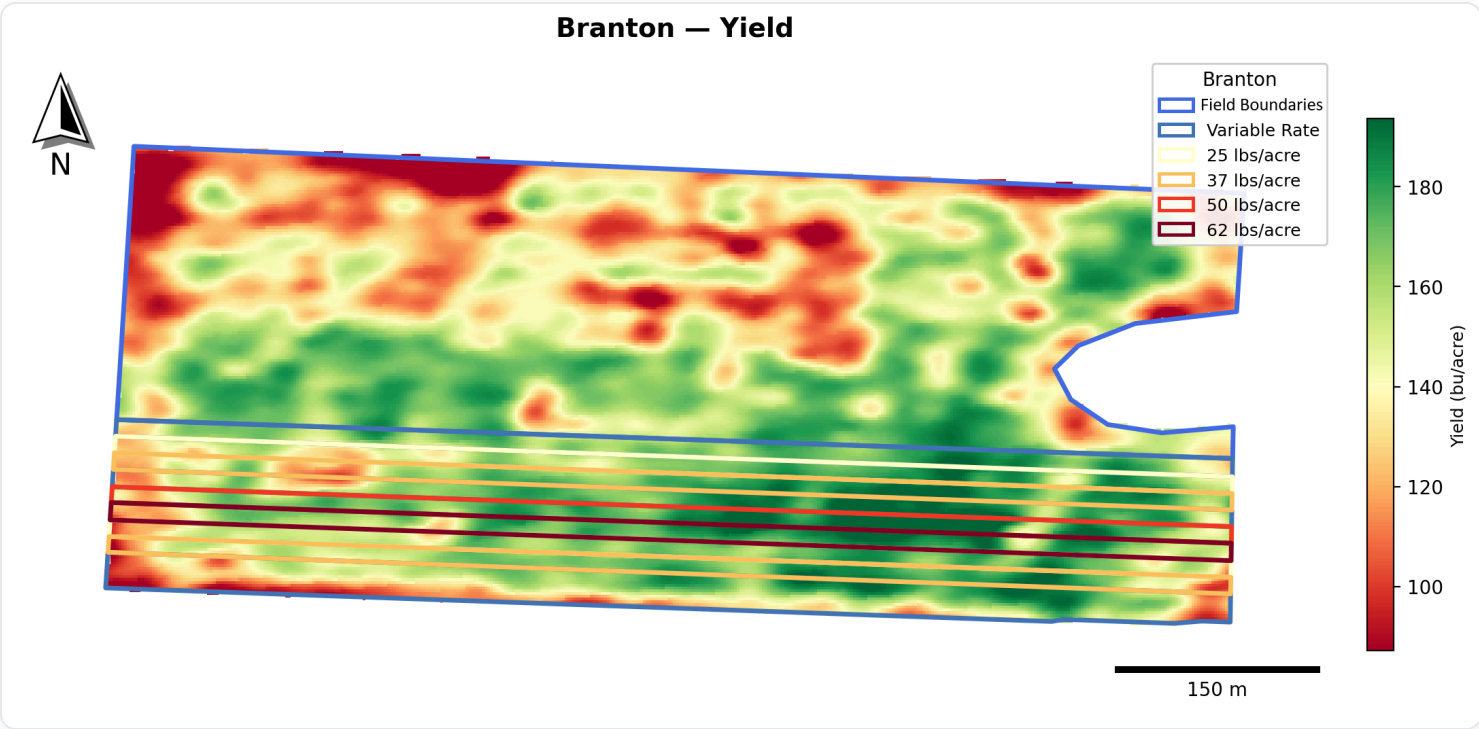


Figure 2. Corn yield spatial distribution.

[Open Interactive NDVI Map](#)

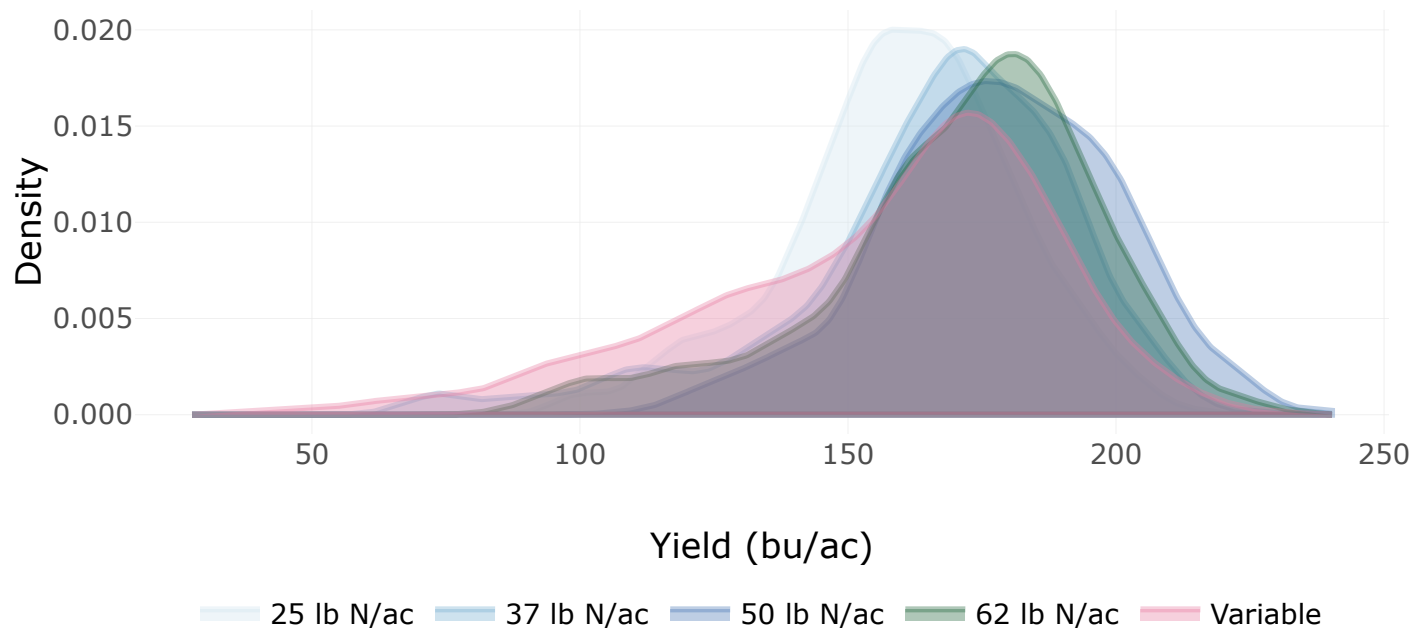
The charts below summarize **yield monitor data** from field 302-3 across the five N rate groups. A small number of extreme outlier points (beyond ± 3 standard deviations within each strip) were removed before analysis to keep the averages clean and representative. Keep in mind that with only one or two strips per fixed rate, we can't run a formal statistical test, but the yield monitor gives us enough data points within each strip to see meaningful patterns.

Density by N Rate Group

Yield Distribution by Strip

Mean $\pm$ SE by N Rate Group

Yield Distribution



Cover Crop Biomass C:N

Cover crop biomass was sampled across field 302-3 in spring **before treatment strips were defined**, all samples represent the field as a whole. The %N of the cover crop residue tells us how nitrogen-rich it is, and the C:N ratio tells us how quickly that nitrogen will be released to the corn crop as the residue breaks down. A C:N ratio below about 25 means decomposition will be relatively fast, and the nitrogen should become available early in the season.

MEAN %N

2.28%

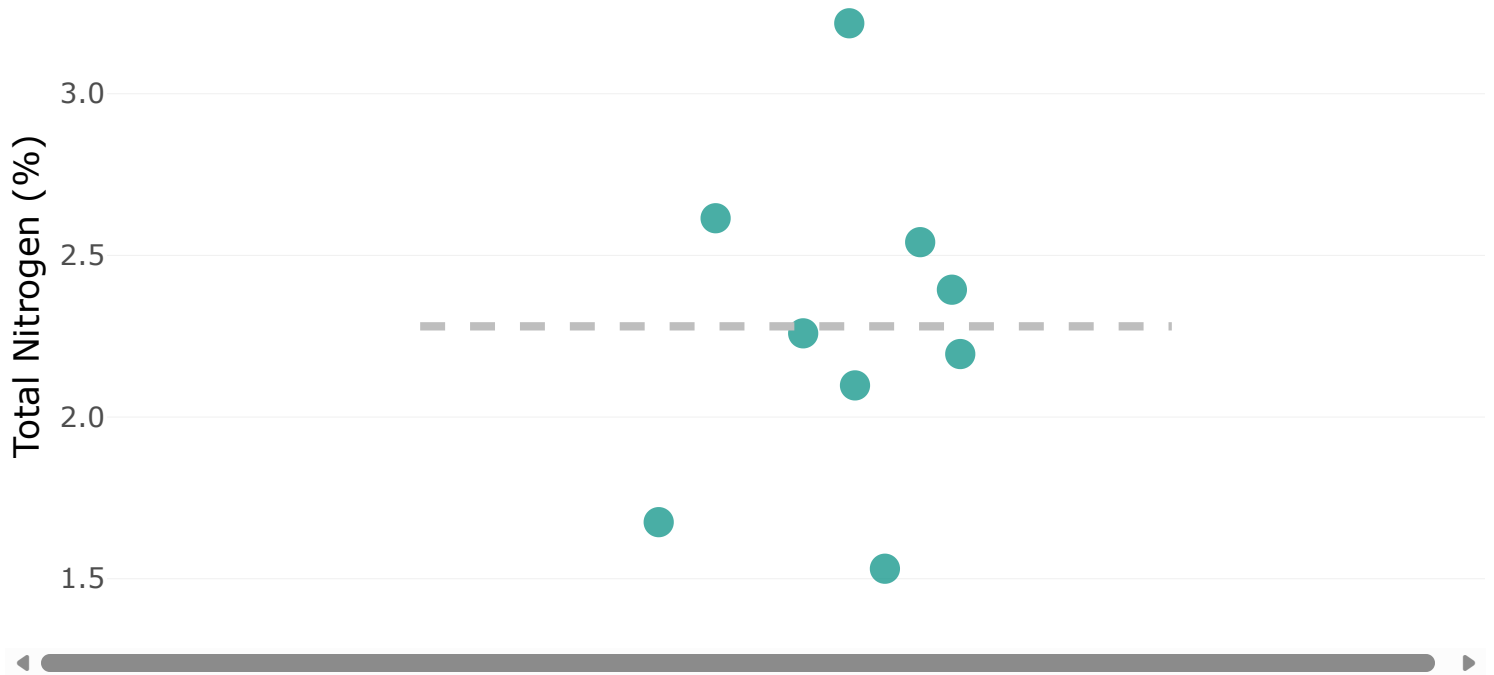
SD = 0.51 | range: 1.53–3.22%

MEAN C:N RATIO

20.6

SD = 5 | range: 14–30

Cover Crop Biomass %N



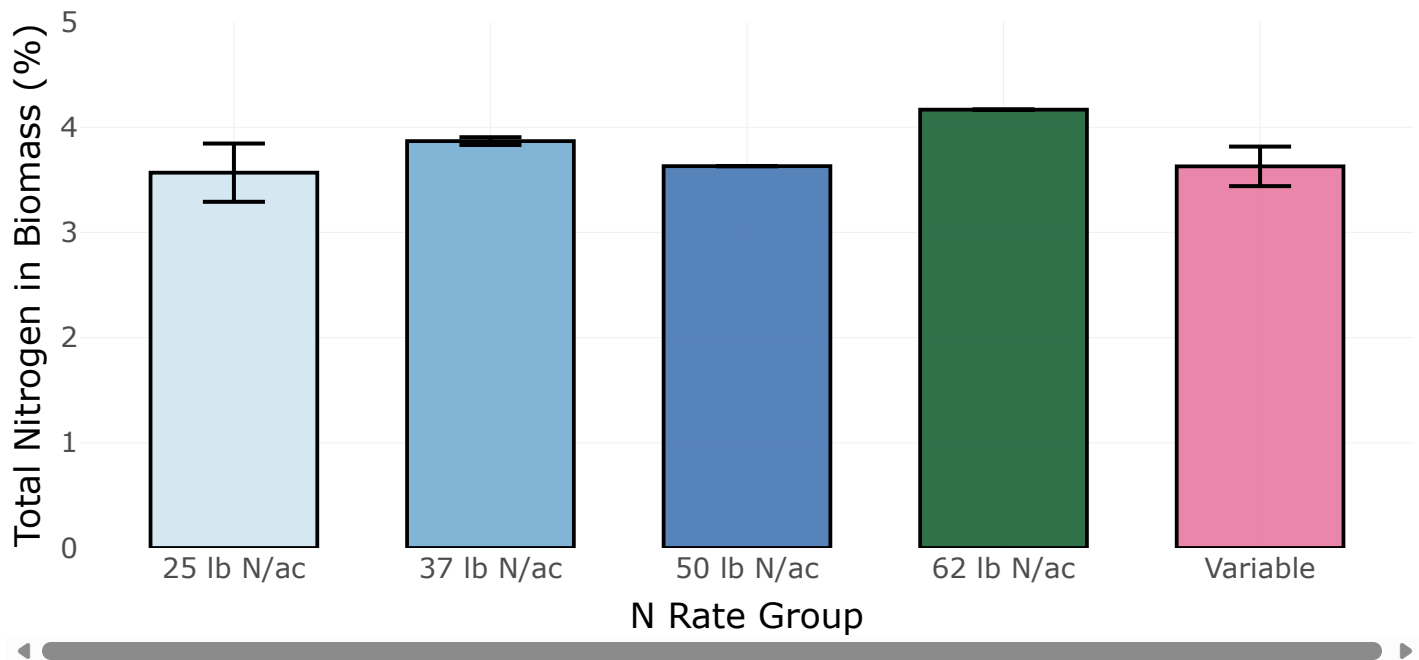
C:N in Corn Biomass

Corn biomass was collected from the 10 N rate strips. Each strip had two sub-samples (A and B); their average is used as the plot value. The bar chart below shows the mean %N $\pm$ SE by N rate group. **Variable-rate strips (R0, R1, R6, R9) are averaged as one group**, allowing comparison with fixed-rate management. Because we have only one or two strips per N rate group, we can't do formal statistics, but the pattern across the groups gives us a useful picture of how nitrogen rate affected what the plant actually took up.

Mean $\pm$ SE by N Rate Group

By Individual Plot

N in Corn Biomass - Mean %N $\pm$ SE by N Rate



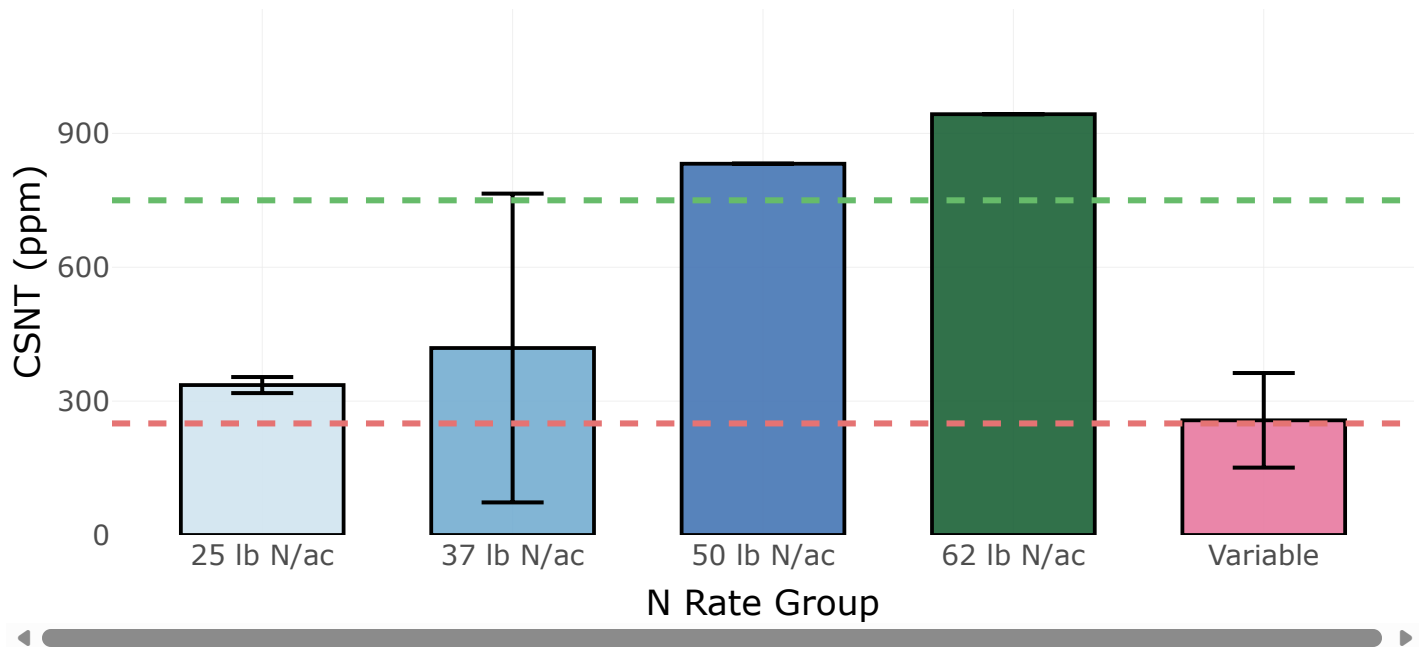
Cornstalk Nitrate Test (CSNT)

The Cornstalk Nitrate Test (CSNT) gives us a post-harvest look at whether the crop had enough nitrogen to carry it all the way through grain fill. It is a post-harvest diagnostic tool: values **below 250 ppm indicate insufficient nitrogen**, while the **optimal range is between 750 and 2000 ppm**. We sampled the lower stalk in two spots per strip, one on the East side and one on the West, and averaged them to get one value per strip.

Mean $\pm$ SE by N Rate Group

[By Individual Strip](#)

Cornstalk Nitrate (CSNT) — Mean $\pm$ SE by N Rate Group



Soil Health Assessment

Soil health was assessed using the **Cornell Soil Health Test**. The overall quality score was **65/100 (High)**. Chemical indicators (pH, P, K, additional nutrients) all scored 100 - ideal. Key opportunities for improvement are in **biological indicators**: Organic matter, soil respiration, and predicted soil protein all scored Low, meaning the microbial community that breaks down residues and cycles nutrients isn't as active as it could be yet. This is where your cover crop system is making its long-term investment, every season of cover cropping adds more carbon and feeds more microbes, gradually building the biological engine that can supply nitrogen from within the soil itself.

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Grower: Donn Branton

Agricultural Service Provider:
Alex Yore
ay327@cornell.edu

Sample ID: QCC94
Field ID: OFE 25 AFTER CC Branton 302-3
Date Sampled: 07/03/2025
Coordinates: Latitude: 42.996261
Longitude: -77.049636

Measured Soil Textural Class: **loam**
Sand: **46%** - Silt: **43%** - Clay: **10%**

Group	Indicator	Value	Rating	Constraints
physical	Predicted Available Water Capacity	0.21	78	
physical	Surface Hardness	179	43	
physical	Subsurface Hardness	245	69	
physical	Aggregate Stability	33.5	55	
biological	Organic Matter Soil Organic Carbon: 1.35 / Total Carbon: 1.4 / Total Nitrogen: 0.13	2.3	22	
biological	Predicted Soil Protein	5.42	37	
biological	Soil Respiration	0.4	22	
biological	Active Carbon	496	49	
chemical	Soil pH	6.6	100	
chemical	Extractable Phosphorus	3.9	100	
chemical	Extractable Potassium	129	100	
chemical	Additional Nutrients Ca: 975.7 / Mg: 110.8 / S: 3.4 Al: 8.1 / B: 0.14 / Cu: 0.16 Fe: 1.6 / Mn: 2.3 / Zn: 0.3		100	

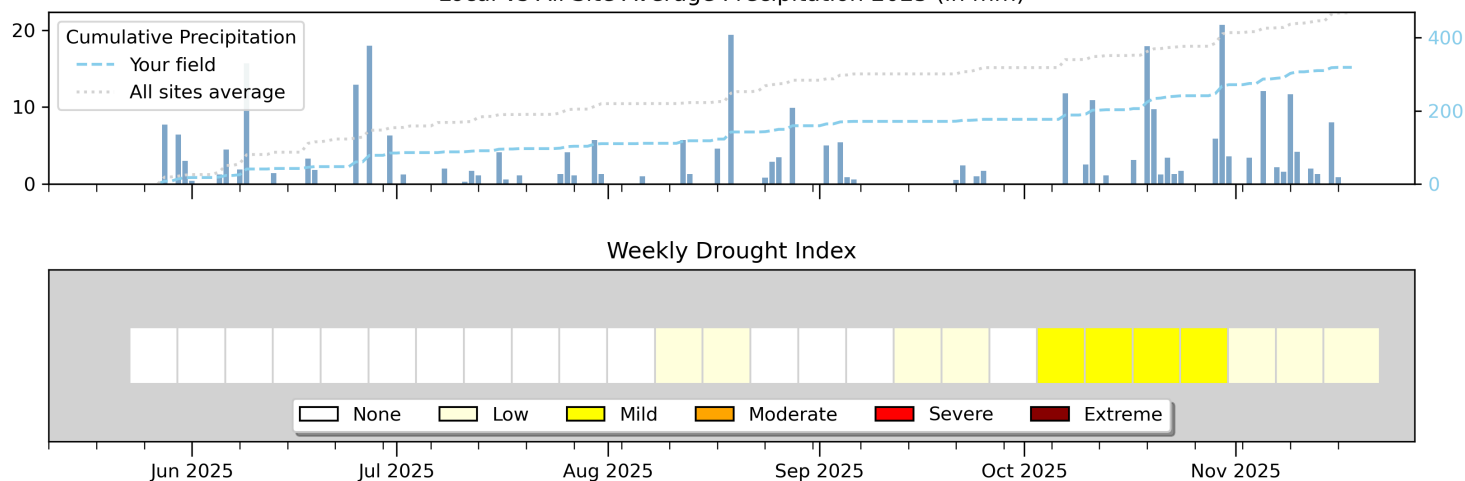
Overall Quality Score: **65** / High

[Open Full Soil Health Report \(PDF\)](#)

Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index, from late May through mid-November 2025. Your field tracked well below the all-site average for cumulative rainfall from June onward, finishing the season around 320 mm (about 13 inches) compared to roughly 470 mm at the network average. Individual rain events were scattered and relatively small, with dry stretches in between; the largest were around 20 mm, in late June, mid-August and late October. The drought index was Low for two weeks in mid-August and two weeks in late September, rose to Mild for most of October, and eased back to Low in November. This drier season is an important backdrop for interpreting your results: drier conditions can limit how much nitrogen the plant can take up from the soil, which may have contributed to the signs of tight nitrogen supply seen in some strips by season end.

Local vs All Site Average Precipitation 2025 (in mm)



Conclusions

COVER CROP BIOMASS

The cover crop came in at an average of **2.28% N** across the field. With a C:N ratio below 25, the residue is in the range where nitrogen should be released relatively quickly during decomposition, meaning some of that N was likely available to the corn early in the season, before fertilizer would have made a difference.

BIOMASS %N IN CORN

Tissue nitrogen ranged from **3.57% at 25 lb N/ac** up to **4.17% at 62 lb N/ac**. The highest rate had the most nitrogen in the plant, but the steps in between were not steady: the 50 lb N/ac strip came in below the 37 lb N/ac strips. With only one or two strips per rate, we cannot confirm this statistically, but the overall direction makes agronomic sense.

CORNSTALK NITRATE (CSNT)

End-of-season stalk nitrate also increased with N rate — from **336 ppm at 25 lb N/ac** up to **943 ppm at 62 lb N/ac**. On average, every fixed-rate group was above the 250 ppm sufficiency threshold, although one of the two 37 lb N/ac strips came in well below it (72 ppm). The variable-rate strips averaged just **257 ppm** — barely above the threshold, suggesting those areas may have been cutting it close on nitrogen supply by harvest.

CORN YIELD

The **50 lb N/ac strip produced the highest yield at 178.1 bu/ac**, outperforming the 62 lb N/ac strip (170.8 bu/ac) and the variable-rate strips (155.7 bu/ac). The yield did not keep climbing with more nitrogen, it peaked at 50 lb N/ac and dropped at the highest rate. That is a useful signal, but each of these rates was a single strip, and two strips that received the same rate (37 lb N/ac) differed by about as much, so it needs to be repeated before settling on a rate.

SOIL HEALTH

The overall soil health score of **65/100 (High)** reflects excellent chemical status but low biological activity. Continued cover cropping could improve organic matter and microbial activity over time, potentially reducing the need for applied N in future seasons.